LocalCloudManager

System Description

Abstract
This document describes the LocalCloudManager support system of the Eclipse Arrowhead framework (documentation edition 5.3.0), as implemented in `ah5-LocalCloudManager-java-spring` of the primary Java source tree (`Eclipse-Arrowhead-java`) (Java-Spring with the vendored Eclipse Arrowhead common library) and deployed with the standalone Compose definition of that folder.

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the LocalCloudManager system. LocalCloudManager is the operator console of the local cloud: it shows the state of the cloud and carries out the management actions an operator starts, with its own Arrowhead identity.

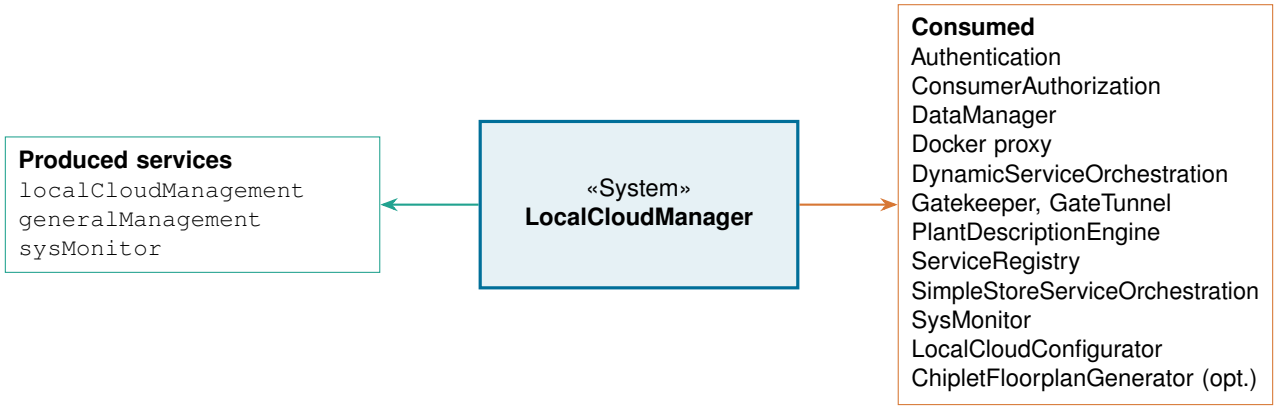


Figure 1: SysML-style block view of LocalCloudManager with its produced and consumed services.

The rest of this document is organized as follows. In Section 2.1, we reference major prior art capabilities of the system. In Section 2.2, we describe the intended usage of the system. In Section 2.3, we describe fundamental properties provided by the system. In Section 2.4, we describe delimitations of capabilities of the system. In Section 3, we describe the abstract service functions consumed or produced by the system. In Section 4, we describe the security capabilities of the system.

2.1 Significant Prior Art

New in this workspace.

2.2 How This System Is Meant to Be Used

The operator opens the console on the console connector (`http://127.0.0.1:8490`), logs in with the console password and works in the views (overview, registry, orchestration rules and authorization policies, plant descriptions, SoS composition, consumers, data, sysMonitor, inter-cloud, runtime and, when enabled, local-cloud configuration and chiplet floorplans).

2.3 System functionalities and properties

2.3.1 Functional properties of the system

- Views assembled from the core and support systems.
- Orchestration rules (create, remove, change priority), plant descriptions and sysMonitor subscriptions.
- Local-cloud management authorization policies of ConsumerAuthorization (view, grant, replace, revoke), with optimistic concurrency on the expected policy snapshot.
- System-of-Systems (SoS) compositions, stored as plant descriptions in the PlantDescriptionEngine; saving an active SoS needs an explicit confirmation because it changes orchestration.
- Optional graphical local-cloud configuration through the LocalCloudConfigurator (catalog, saved designs, plans and reviewed deployments), consumed as the Arrowhead service `localCloudConfiguration` through orchestration; switched off with `lcm.local-cloud-configuration.enabled=false`.
- Optional chiplet floorplan editing through the ChipletFloorplanGenerator (orchestrated) when `lcm.chiplet-floorplans.enabled=true`.
- Container list, logs and lifecycle through the separate Docker proxy.
- The same views and actions for Arrowhead systems on the Arrowhead connector (`localCloudManagement`), for Sysop, whitelisted systems and systems with a ConsumerAuthorization grant.
- Native sysMonitor observations (common publisher) and `generalManagement get-config`.

2.3.2 Configuration of system properties

The system is a Spring Boot application configured with Spring properties: the packaged `application.properties` is a development sample, the deployment configuration `config/application.properties` maps the Compose environment variables to them. The properties that shape its behaviour are:

Property	Deployment value	Meaning
`server.port`	8491	Arrowhead connector (mutual TLS).
`console.port` / `console.address`	8490 / 0.0.0.0 in the container	Console connector; Compose publishes it on the host's loopback interface only.
`server.ssl.*`	PKCS12 key store and truststore, `client-auth=need`	System certificate `LocalCloudManager.<Cloud>.<Organization>.arrowhead.eu`.

Property	Deployment value	Meaning
`authentication.policy`	`outsourced`	Identity tokens of Authentication; own login with `authenticator.credentials` (secret `ah.identity-password`).
`service.interface.policy`	`CERT_AUTH`	Policy of the registered interfaces.
`management.policy`	`sysop-only`	`generalManagement`.
`service.authorization.enabled`	true	ConsumerAuthorization check of `localCloudManagement` operations.
`lcm.api.whitelist`	empty	Systems that need no grant.
`arrowhead.ssl.crl-file`	`/crl/ca.crl`	CRL of the cloud CA (provider certificates; remaining validity in the overview).
`http.collector.mode`, `http.collector.orchestration.ttl`	`ORCHESTRATION`, 300	Providers selected by orchestration; reuse of results in seconds.
`console.password.hash`	secret `ah.console-password-hash`	`scrypt:<salt>:<hash>`; without it the console answers 503.
`console.allowed.hosts`	127.0.0.1, localhost, [::1]	Accepted Host and Origin names.
`console.proxy-secret`	empty	Optional: only requests with `X-Console-Proxy` are served (behind a reverse proxy).
`console.session.seconds`	28800	Console session lifetime.
`lcm.compose-project`, `lcm.docker.url`	managed project, `http://docker-proxy:2375`	Compose project shown; Docker proxy.
`lcm.managed-services`	23 services by default	Containers the console may start, stop and restart.
`lcm.local-cloud-configuration.enabled`	true	Local-cloud configuration view.
`lcm.chiplet-floorplans.enabled`, `lcm.chiplet-manager`	false	Chiplet floorplan view; console name ChipletLocalCloudManager.
`lcm.orchestration-priority-mode`	`native`	`replace`: change a rule priority by create and remove instead of SimpleStore `modify-priorities`.
`sysmonitor.*`	enabled, `amqp-broker:5671`	Common sysMonitor publisher.
`arrowhead.security.profile`	`secure` (default)	The security profile guard refuses to start with TLS off, declared identities or `service.authorization.enabled=false`; `development` only explicitly, for isolated tests.

Secrets (store password, identity password, console password hash) have no usable defaults; they are Compose file secrets read through `configtree:/run/secrets/`.

2.3.3 Data stored by the system

None durable: the system has no database. Console sessions, the identity token, orchestration results and the overview history are kept in memory. SoS compositions are stored by the PlantDescriptionEngine, policies by ConsumerAuthorization and designs by the LocalCloudConfigurator.

2.3.4 Non functional properties

A failing provider gives a partial view with an error note.

2.3.5 Stateful or stateless

Stateless apart from sessions.

2.4 Important Delimitations

Acts only within the grants given to LocalCloudManager (not a sysop since J23). The console connector must not be exposed beyond the host: it is plain HTTP, authenticated by the console password only. Docker is reached only through the proxy, whose allowlist limits the console to listing, logs and the lifecycle of managed services of its own Compose project.

3 Services

This section describes the produced and consumed services. Each produced service is specified by an SD document and one IDD document per interface.

3.1 Produced services

Service	Version	Interfaces	SD	IDD
`localCloudManagement`	1.0.0	HTTP(S)	[1]	[2]
`sysMonitor`	1.0.0	AMQP(S)	[3]	[4]
`generalManagement`	1.0.0	HTTP(S)	–	–

`localCloudManagement` is registered with `generic_https` and policy `CERT_AUTH` on the Arrowhead connector (8491). The console-only operations (session, login, logout) and `GET /health` exist on the console connector only; the console routes added after the registration list (features, authorization policies, rule priority, SoS, local-cloud configuration, chiplet floorplans) are served on both connectors but not registered. `general Management` offers `get-config` only (no database, so no `get-log`). The IDD specifies all routes.

3.2 Consumed services

Provider	Service	Operations	Use	Condition
ServiceRegistry	systemDiscovery, serviceDiscovery; serviceRegistry Management	register, revoke, lookup; device, system and service-instance queries	Registration (common listener); inventory.	always
Authentication	identity	identity-login, identity-verify, identity-logout	Own token (common login job); verification of caller tokens on 8491.	always
DynamicService Orchestration	service Orchestration; service Orchestration History Management	pull; query	Orchestration of all other services; history.	always
SimpleStore Service Orchestration	service Orchestration SimpleStore Management	query, create, remove, modify-priorities	Rules and rule priorities.	grant
Consumer Authorization	authorization Management	query-policies, grant-policies, revoke-policies	Consumers view; policy editor.	grant
DataManager	dataHistorian	list-systems, get-data	Data view.	grant
SysMonitor	sysMonitorHistory; sysMonitor Management	all	sysMonitor views.	grant

Provider	Service	Operations	Use	Condition
PlantDescription Engine	plantDescription Management	all	PD and SoS views.	PDE whitelist with LocalCloudManager
Gatekeeper, GateTunnel	gatekeeper Management; gate TunnelManagement	status, sessions	Inter-cloud view.	grant
Docker proxy	Docker Engine subset	list, logs, start/stop/restart	Runtime view, sysMonitor health.	always (internal network)
LocalCloud Configurator	localCloud Configuration	catalog, status, plan, submit-deployment, get-deployment, list-, create-, get-, update-, remove-configuration	Local-cloud configuration view.	view enabled, grant
Consumer Authorization	authorization	verify	Grants of callers of `localCloud Management` on 8491 (cached 30 s).	always
ChipletFloorplan Generator	floorplanDiscovery; floorplan Management	query-floorplans, get-floorplan; create, replace, remove, check-network	Floorplan view.	floorplans enabled, grant

The common service collector runs in `ORCHESTRATION` mode: only the bootstrap services (ServiceRegistry, Authentication `identity`, the orchestrator) are looked up in the ServiceRegistry; every other Arrowhead service is selected by an orchestration request on behalf of LocalCloudManager (results reused for 300 s), and every such call needs a ConsumerAuthorization grant.

4 Security

LocalCloudManager is a Java-Spring support system with two connectors in one process. The Arrowhead connector (8491) uses HTTPS with mutual TLS and the X.509 certificate following the Arrowhead certificate profile (`LocalCloudManager.<Cloud>.<Organization>.arrowhead.eu`). Towards providers it uses the same certificate and an identity token bound to it; provider certificates not issued under the cloud CA, or revoked in its CRL, are refused. The console connector (8490) is plain HTTP, published on the host's loopback address only, for the operator's browser. The standalone deployment of this folder (`compose.yaml`) runs this configuration.

4.1 Security model

- **Protocols:** HTTPS (TLS 1.3 and 1.2) with mutual TLS on 8491 and towards the cloud; AMQP over TLS (SASL EXTERNAL) for `sysMonitor`; plain HTTP on the console connector (host loopback) and between the console process and the Docker proxy (internal network).
- **Security profile:** the common security profile guard refuses to start the system with TLS disabled, declared identities or the authorization check switched off, unless the `development` profile is selected explicitly.
- **Data protection:** transport encryption by TLS towards the cloud; no database; secrets as Compose file secrets, never in the environment or on the command line; read-only root filesystem, UID 10001, no capabilities.
- **Arrowhead callers (8491):** client certificate and identity token of the same system (outsourced authentication); `localCloudManagement` for Sysop, `lcm.api.whitelist` and systems with a Consumer-Authorization grant of the operation (fail closed); `generalManagement` for Sysop only.
- **Operator (8490):** console password (scrypt hash, N=16384, r=8, p=1), JSON-only login, HttpOnly SameSite=Strict session cookie checked by Spring Security (stateless), per-client lock-out after five wrong passwords within five minutes (60 s, doubling to at most 15 minutes), allowed Host and Origin names (against DNS rebinding), optional proxy secret, `X-Arrowhead-Action: confirm` on changes and expected-snapshot checks (409) for policy, rule-priority, SoS and floorplan edits. Console requests bypass the Arrowhead filters; they act with LocalCloudManager's own grants.
- **Privilege separation:** the console process holds no Docker socket. The separate Docker proxy forwards only `GET /containers/json` (filtered to its Compose project), `GET /containers/<id>/logs` of a project container and `POST /containers/<id>/start|stop|restart` of a project container in `MANAGED_SERVICES`; everything else is answered 403.

For the Arrowhead certificate profile see github.com/eclipse-arrowhead/documentation.

5 References

[1] J. Delsing, "localCloudManagement – Service Description (LocalCloudManager)," Eclipse Arrowhead, SD 5.3.0, 2026, localCloudManager-v5.3.0-localCloudManagement-SD.pdf.

[2] ——, "localCloudManagement HTTP/TLS/JSON – Interface Design Description (LocalCloudManager)," Eclipse Arrowhead, IDD 5.3.0, 2026, localCloudManager-v5.3.0-localCloudManagement-HTTP-IDD.pdf.

[3] ——, "sysMonitor – Service Description (LocalCloudManager)," Eclipse Arrowhead, SD 5.3.0, 2026, localCloudManager-v5.3.0-sysMonitor-SD.pdf.

[4] ——, "sysMonitor AMQP/TLS/JSON – Interface Design Description (LocalCloudManager)," Eclipse Arrowhead, IDD 5.3.0, 2026, localCloudManager-v5.3.0-sysMonitor-AMQP-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): console functions, configuration, consumed services and security updated to the current Node.js implementation (policies, rule priority, SoS, local-cloud configuration, chiplet floorplans).	Claude Opus 5.5 (draft)
3	2026-11-09	5.2.2	Java-Spring implementation: Spring Boot with the vendored common library and no database; Arrowhead connector 8491 (mutual TLS, identity token, `generic_https/CERT_AUTH`) and console connector 8490 (password login in Spring Security); separate Docker proxy; LocalCloudConfigurator consumed through orchestration; `generalManagement`; security profile guard. Node.js/nginx descriptions removed.	Claude Opus 5.5 (draft)
4	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	